MTL106: Probability and Stochastic Processes

Solutions of Practice Sheet 2

1. Let A be the event that a gadget is produced by Machine A, B be the event that it is produced by Machine B, and H be the event that the gadget is of high quality.

   Given

   $$P(A) = 0.70, \quad P(B) = 0.30, \quad P(H|A) = 0.95, \quad P(H|B) = 0.90.$$

   We have to calculate $P(A|H)$. By Bayes' theorem,

   $$P(A|H) = \frac{P(H|A)P(A)}{P(H)},$$

   where

   $$P(H) = P(H|A)P(A) + P(H|B)P(B).$$

2. Given

   $$P(E_1) = P(E_2) = P(E_3) = \frac{1}{2},$$

   $$P(E_1E_2) = P(E_2E_3) = P(E_1E_3) = \frac{1}{4}, \quad P(E_1E_2E_3) = \frac{1}{4}.$$

   Hence, the events E_1, E_2, E_3 are not mutually independent, but they are pairwise independent.

3. **Inverse Images**
   Solution:

   (a) $x = 0.5$: Only $X(\omega_4) = 0 \leq 0.5$. Set: $\{\omega_4\}$.

   (b) $x = 1.5$: $X(\omega) \in \{0, 1\}$. Set: $\{\omega_1, \omega_2, \omega_4\}$.

   (c) $x = 2.5$: All values ≤ 2.5. Set: Ω.

   Since $\mathcal{F}$ is the power set, all subsets are events. Thus X is a random variable.

4. **Binary Strings**
   Solution: (a) S is the set of all binary tuples of length 50. Cardinality $|S| = 2^{50}$.
   (b) Support of X: $\{0, 1, 2, \ldots, 50\}$.
   (c) $P(X = i) = \frac{\binom{50}{i}}{2^{50}} = \binom{50}{i}(0.5)^{50}$.
   (d) $P(X = 27) = \binom{50}{27}(0.5)^{50}$.

5. **Pairwise vs. Mutual Independence**
   Solution: The sample space is $\Omega = \{HH, HT, TH, TT\}$, with each outcome having probability $1/4$.

- $P(E_1) = P(\{HH, HT\}) = 2/4 = 1/2$.
- $P(E_2) = P(\{HH, TH\}) = 2/4 = 1/2$.
- $P(E_3) = P(\{HT, TH\}) = 2/4 = 1/2$.

Pairwise Independence:

- $P(E_1 \cap E_2) = P(\{HH\}) = 1/4$. Check: $P(E_1)P(E_2) = (1/2)(1/2) = 1/4$. (OK)
- $P(E_1 \cap E_3) = P(\{HT\}) = 1/4$. Check: $P(E_1)P(E_3) = (1/2)(1/2) = 1/4$. (OK)
- $P(E_2 \cap E_3) = P(\{TH\}) = 1/4$. Check: $P(E_2)P(E_3) = (1/2)(1/2) = 1/4$. (OK)

Thus, they are pairwise independent.
Mutual Independence: Consider $P(E_1 \cap E_2 \cap E_3)$. The intersection is empty because one cannot have "Exactly one Head" if both tosses are Heads.

$$P(E_1 \cap E_2 \cap E_3) = P(\emptyset) = 0$$

However, $P(E_1)P(E_2)P(E_3) = (1/2)^3 = 1/8$. Since $0 \neq 1/8$, the events are **not** mutually independent.

6. **CDF Analysis (Mixed Type)**
Solution:

(a) **Properties Check:**

- **Limits:** $\lim_{x\to-\infty} F(x) = 0$ and $\lim_{x\to\infty} F(x) = 1$. (Satisfied)
- **Non-decreasing:** The function segments 0, $x^2/4$, $1/2$ and 1 are non-decreasing in their respective intervals. At the boundaries:
  - At $x = 1$: Left limit $1^2/4 = 0.25 < 0.5$ (Jump up).
  - At $x = 2$: Left limit $0.5 < 1$ (Jump up).

  (Satisfied)
- **Right Continuity:** $\lim_{h\to0} F(x + h) = F(x)$. The function is defined using $\leq$ on the left of intervals and $\geq$ on the right, matching the standard $P(X \leq x)$ definition. Specifically, at jumps $x = 1$ and $x = 2$, the value equals the limit from the right. (Satisfied)

(b) **Discontinuities:** The jumps occur where the definition changes abruptly.

- At $x = 1$: $F(1) = 0.5$, $F(1^-) = 1^2/4 = 0.25$. Mass $P(X = 1) = 0.5 - 0.25 = 0.25$.
- At $x = 2$: $F(2) = 1$, $F(2^-) = 0.5$. Mass $P(X = 2) = 1 - 0.5 = 0.5$.

(c) $P(0.5 < X \leq 1.5) = F(1.5) - F(0.5)$.

- $F(1.5) = 0.5$ (since $1 \leq 1.5 < 2$).
- $F(0.5) = (0.5)^2/4 = 0.0625$ (since $0 \leq 0.5 < 1$).

Result: $0.5 - 0.0625 = 0.4375$.

7. **PMF Constants**
Solution:

(a) Sum of probabilities must be 1:

$$k(0.5)^1 + k(0.5)^2 + k(0.5)^3 + 4k = 1.$$

$$k(0.5 + 0.25 + 0.125 + 4) = 1 \implies k(4.875) = 1.$$

$$k = \frac{1}{4.875} = \frac{8}{39}.$$

(b) **CDF Construction:** The probabilities are: $P(1) = \frac{4}{39}$, $P(2) = \frac{2}{39}$, $P(3) = \frac{1}{39}$, $P(4) = \frac{32}{39}$.

$$F(x) = \begin{cases} 0 & x < 1 \\ 4/39 & 1 \leq x < 2 \\ 6/39 & 2 \leq x < 3 \\ 7/39 & 3 \leq x < 4 \\ 1 & x \geq 4 \end{cases}$$

8. **River flow**
Solution:

(a) **Verify:** $F(1) = 1 - 1 = 0$. $\lim_{y\to\infty}(1 - 0) = 1$. Derivative $2y^{-3} > 0$ for $y \geq 1$. Valid.

(b) **PDF:**

$$f_Y(y) = \frac{d}{dy}\left(1 - y^{-2}\right) = 2y^{-3} = \frac{2}{y^3}, \quad y \geq 1.$$

(c) **Transformation:** $X = k(Y - 1)$.

$$F_X(x) = P(X \leq x) = P(k(Y-1) \leq x) = P\left(Y \leq \frac{x}{k} + 1\right).$$

Substitute $y = \frac{x}{k} + 1$ into $F_Y(y)$:

$$F_X(x) = 1 - \frac{1}{(\frac{x}{k} + 1)^2} = 1 - \frac{1}{(\frac{x+k}{k})^2} = 1 - \frac{k^2}{(x+k)^2}, \quad x \geq 0.$$